

DCAGTN: Dual-snapshot causal-attention graph transformer with feedback mechanism for real-time rumor detection

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ABSTRACT

The rapid spread of rumors on social media has become a global concern, fueling misinformation, harming reputations, and disrupting societies. Despite recent progress, existing rumor-detection methods still face major limitations. Static graph models overlook the evolving nature of online interactions, require costly recomputation, and struggle to adapt. Many dynamic models, meanwhile, fail to preserve chronological consistency, leading to temporal leakage and weaker early-stage predictions. The constant flow of user-generated content also adds computational demands, making scalability and timely response difficult. Moreover, most methods focus narrowly on temporal or structural aspects while neglecting causal-consistency modeling, which limits interpretability and their ability to explain how rumors spread. Addressing these challenges requires approaches that jointly model both the semantic content of posts and the relational dynamics (replies, shares, mentions, endorsements) that shape information flow across networks. To this end, we propose the Dynamic Causal-Attention Graph Transformer Network (**DCAGTN**), an intelligent framework integrating *sequential and temporal snapshots* with *causal attention*, *feedback refinement*, and *causal alignment* for dynamic rumor detection. Specifically: (1) a *dual-snapshot mechanism* captures short-term interactions and long-term propagation; (2) a *causal-consistency attention mechanism* prioritizes time-sensitive nodes by enforcing one-way chronological information flow, preserving event chronology and eliminating temporal leakage; (3) a lightweight *feedback mechanism* incrementally refines node embeddings without full graph recomputation; and (4) a *causal alignment loss* enforces temporal consistency for robust and interpretable decision-making. Extensive experiments on *Twitter15*, *Twitter16*, and *Weibo* confirm the effectiveness of DCAGTN, achieving state-of-the-art accuracy, efficient early detection, and enhanced causal interpretability.

1. Introduction

Social media platforms are an essential tool for communication by billions of people worldwide in real-time.¹ Their rapid growth has provided unprecedented opportunities for the production, dissemination, and consumption of rumors and misinformation [1]. Such rumors distort public understanding [2], influence individual and collective behavior [3,4], and erode trust in societal institutions.² In practice, this requires capturing not only the semantic content of posts but also the structural interactions through which they spread (replies, shares, likes,

mentions, and endorsements), which jointly shape influence flow across networks [5]. Hence, effective rumor detection methods are crucial for public access to reliable information.

The goal of rumor detection is to identify and verify the truthfulness of information shared [6]. However, this task remains challenging due to several persistent limitations in existing models. *First*, static or uniformly weighted graph formulations are ill-suited to real-time streams, missing the rapidly evolving propagation structure and often requiring costly recomputation when new data arrives [7–10]. *Second*, models that update over time struggle to preserve chronological order,

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¹ <https://www.statista.com/statistics/278414/number-of-worldwide-social-network-users/>

² <https://www.nytimes.com/2025/03/24/business/trump-misinformation-false-claims.html>

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causing temporal leakage and degrading early-stage predictions. *Third*, the massive and continuous influx of user-generated content imposes significant computational burdens, making it difficult for detection systems to process high-frequency data streams without delay [11]. *Finally*, most existing rumor detection models focus on temporal or structural features but overlook *causal-consistency reasoning*, limiting their ability to explain how chronological dependencies drive rumor spread and how interventions might suppress misinformation diffusion [12].

Recent rumor detection research has attempted to address these challenges, but existing solutions remain incomplete. Static graph approaches [7–10,13–15] are effective for historical analysis yet fail to adapt to the evolving structure of real-time social media streams. Dynamic models such as Dynamic GCN [16], DGNF [17], and DDGCN [18] capture temporal evolution but often lose chronological consistency during early-stage propagation [19]. Attention-based models, including GAT [20], its recent variants [21], and transformer-based architectures [22], enhance representation learning but suffer from temporal misalignment and limited adaptability in real-time environments. While some methods integrate time-aware components [23] to manage large-scale data streams, many still depend on costly sequential or full-graph recomputation. Moreover, causal-consistency modeling, essential for explaining why certain posts gain traction, remains underutilized [12,19,24]. Even recent state-of-the-art models such as FGDGNN [25] and PSGT [26] improve temporal modeling and structure-aware attention, yet still lack explicit causal supervision and lightweight feedback mechanisms for efficient adaptation.

In this paper, to address the challenges, we propose a novel dual-snapshot graph-based model, termed Dynamic Causal-Attention Graph-Transformer Network (**DCAGTN**). Specifically, (1) To overcome the limitations of static modeling, we employ a *dual-snapshot mechanism* that captures both sequential and temporal snapshots of the propagation network. These snapshots are processed by a dynamic graph attention variant (GATv2Conv), which is strictly more expressive than GAT and allows the model to adapt continuously to evolving interactions. (2) To address the challenge of modeling time-sensitive nodes, we introduce a *causal attention layer* that dynamically prioritizes these nodes, ensuring temporal consistency in representation learning. By leveraging temporal ordering, the layer enhances early-stage rumor detection and prevents information leakage from future events into current node representations, thereby preserving causal integrity in the model's predictions. (3) To balance computational efficiency with detection accuracy, we incorporate a *Feedback Mechanism* that iteratively refines node embeddings without requiring full graph re-computation. This mechanism updates embeddings based on feedback from previous predictions, enabling rapid adaptation to new information and significantly reducing computational burden. (4) To solve the problem of missing causal-consistency modeling, we leverage a *Causal Alignment Loss* that enforces temporal coherence during training, enabling the model to distinguish causally significant patterns from noise. This reduces misclassification due to temporal misalignment and ensures more robust and interpretable decision-making processes. We summarize our contributions as follows:

- We propose *DCAGTN*, a dual-stage dynamic graph framework that integrates causal attention and graph adaptation to capture evolving rumor dynamics. The model processes both sequential and temporal snapshots to track short-term interactions and long-term propagation, enabling accurate real-time rumor detection.
- We introduce a *causal attention mechanism* that identifies and prioritizes time-sensitive nodes. This module improves early detection by aligning node importance with temporal order during rumor propagation.
- To enhance adaptability and reduce computational cost, we design a *dual-snapshot* using GATv2Conv and a lightweight *feedback mechanism* that updates embeddings incrementally without full graph recomputation.

- We propose a *causal alignment loss* that supervises temporal consistency during training, allowing the model to learn causally meaningful patterns and improve interpretability.
- Extensive experiments on benchmark datasets demonstrate that DCAGTN achieves state-of-the-art accuracy and efficiency, outperforming prior models in both early-stage and real-time rumor detection situations.

The rest of our work is organized as follows: [Section 2](#) reviews the related work on rumor detection. [Section 3](#) discusses our model methodology. [Section 4](#) analyses relevant experiments and evaluations. [Section 5](#) provides the conclusions of our work.

2. Related work

In this section, we discuss related work on methodological advances and persistent limitations in rumor detection and dynamic graph modeling, anchored by recent surveys on dynamic GNNs [27] and rumor detection [28].

2.1. Traditional methods

Early rumor-detection studies relied heavily on handcrafted features and classical classifiers, which provided valuable early insights but lacked adaptability to large-scale, rapidly evolving social media platforms. Castillo et al. [29] introduced a Decision Tree Classifier (DTC) using surface-level features such as punctuation counts, URLs, and tweet metadata. Qazvinian et al. [30] expanded feature scope to include user credibility and topic relevance, while Yang et al. [31] integrated geolocation and client-based attributes. These methods, although computationally lightweight, were fundamentally constrained by their reliance on static, manually engineered features, making them unsuitable for real-time rumor streams where the structural and semantic context evolves rapidly. Later efforts, such as TDRD [32], exploited source-level features for near-real-time detection, and Liu et al. [33] combined RNN and CNN modules to capture sequential user interactions. Nonetheless, these systems still fell short in modeling the dynamic structure of social networks, highlighting the need for models that can learn from evolving relational contexts.

2.2. Graph-based and NLP approaches

The limitations of traditional methods motivated the adoption of deep learning, particularly in Natural Language Processing (NLP) and graph representation learning, to capture richer structural and temporal patterns in rumor propagation. Sequence-based NLP models such as SVM-TK [34] and RvNN [35] introduced tree-structured architectures to represent source-reply hierarchies. Bian et al. [13] advanced this with BiGCN, which processes both upstream and downstream propagation directions to better encode information flow. Graph embedding techniques such as DeepWalk [7] and Node2Vec [8] provided unsupervised node representations but were inherently static, failing to adapt to evolving network states.

Dynamic graph models emerged to address temporal variation. D_{YN}GCN [16] aggregates sequential snapshots, DGNF [17] models discrete-time changes, and DA-GCN [36] integrates dual-attention to filter irrelevant interactions. These approaches improved adaptability over static embeddings, but they primarily focus on temporal aggregation rather than explicit causal constraints, leaving room for further advances in temporal consistency and interpretability.

2.3. Graph attention networks and transformer-based mechanisms

The integration of attention mechanisms marked a major shift toward more adaptive rumor detection. Transformers, introduced by Vaswani et al. [22], demonstrated exceptional capability in capturing long-range

dependencies and contextual relationships, making them well-suited for modeling both semantic and temporal dynamics in social media posts. Ma et al. [37] applied hierarchical attention to source-reply interactions, while Ran et al. [23] proposed multi-channel graph attention networks (MGAT) to encode semantic and relational features simultaneously.

Graph Attention Networks (GAT) [20] extended attention mechanisms to graph structures by learning adaptive neighbor weights, offering improved discriminative power over GCNs. Building on this foundation, recent baselines such as AGAD [21], PSGT [26], and FGDGNN [25] further illustrate the field's shift toward fine-grained dynamic attention through adversarial training, progressive semantic guidance, and adaptive temporal filtering. While these models advance robustness and adaptability, they remain attention-driven rather than causally grounded, motivating complementary approaches such as DCAGTN.

Beyond graph-centric designs, recent research has leveraged large language models (LLMs) such as transformer-based and GPT-style architectures for text-oriented rumor detection and stance classification [38–41]. These models achieve strong semantic and contextual understanding in low-structure or text-dominant settings but often overlook relational and temporal propagation cues. DCAGTN complements these efforts by embedding causal-consistency modeling and feedback refinement into dynamic graph representations, enabling context-aware and temporally coherent detection even as propagation structures evolve over time.

2.4. Dynamic causal-attention methods

Causal-consistency modeling has recently emerged as a promising tool to bridge the gap between temporal adaptation and interpretability in rumor detection. Wu et al. [42] proposed PGNN, a gated GNN for temporal propagation, while Rao et al. [43] addressed cross-modal co-attention alignment in dynamic graphs. DDGCN [18] combined dual-dynamic graph convolutions to handle structural changes but lacked a causal alignment strategy to prioritize intervention-relevant nodes. Cheung et al. [19] introduced CDGNN, explicitly integrating causal attention to improve generalization to unseen rumor patterns. Kwao et al. [24] extended causal-consistency modeling to multimodal misinformation detection, focusing on quantifying misinformation drivers rather than embedding causal constraints into propagation models. In parallel, recent attention-driven baselines such as AGAD [21], PSGT [26], and FGDGNN [25] introduced adversarial training, progressive semantic guidance, and adaptive temporal filtering, but they do not incorporate explicit causal supervision or feedback refinement. General dynamic graph frameworks such as UniDyG [44] unify discrete- and continuous-time graphs via temporal interpolation to improve the scalability of graph representations. In contrast, DCAGTN explicitly separates *sequential* and *temporal* snapshots and optimizes them under a *causal-consistency* objective that enforces one-way chronological flow. This separation preserves fine-grained early diffusion cues that interpolation can smooth out, which is crucial for real-time rumor detection.

Despite these advances, no existing approach fully unifies time-sensitive learning with causal-consistency modeling in a lightweight, feedback-enabled model that can continuously adapt as rumors spread. This motivates our proposed DCAGTN, which explicitly integrates dual-snapshot temporal modeling, causal attention, and incremental feedback mechanisms, a combination not present in prior baselines. In the next section, we describe the model and its implementation.

3. Method

Problem Definition and Notations. We jointly model the semantic content of posts and the evolving propagation patterns as information spreads across user interaction networks [6]. Formally, we define a dataset D comprising event instances $C = \{C_1, C_2, \dots, C_N\}$, where each C_i is a conversation thread with a source post s_i and its replies or

retweets $\xi = \{r_{ij}\}$. The relational structure is a graph $G = (V, E)$ whose nodes V are posts and edges E are interactions (reply/retweet/mention). The adjacency matrix $A \in \{0, 1\}^{|V| \times |V|}$ is

$$A_{ij} = \begin{cases} 1 & \text{if post } i \text{ links to } j \text{ (reply, retweet, mention)} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Each node $v_i \in V$ has (i) a textual feature $T_i \in \mathbb{R}^{d_t}$ and (ii) a structural feature $S_i \in \mathbb{R}^{d_s}$. Text reflects *what* is said; structure reflects *how* it spreads [13,45]. Our goal is to learn a mapping $f : (T_i, S_i, A, t) \rightarrow L_i$, where $L_i \in \{NR, FR, TR, UR\}$ denotes the veracity label.

Key notations are as follows:

- T_i : Textual embedding of post i , obtained via a pretrained RoBERTa encoder.
- S_i : Structural feature vector of post i , incorporating time, edge type, and local graph properties.
- R : Set of relation types (e.g., reply, retweet, mention) used to define edge semantics.
- A : Adjacency matrix capturing directed interactions among posts.

3.1. Model overview

In this work, we propose **DCAGTN**, a dual-snapshot graph-based framework for real-time rumor detection on social media. As illustrated in Fig. 1, DCAGTN integrates five key components into a unified end-to-end framework. *First, Node representations* are initialized by combining RoBERTa-based contextual embeddings with structural and relational features from social interactions. *Second, Dual graph snapshots* are constructed to capture rumor propagation at different timescales: sequential snapshots $S(t)$ model fine-grained diffusion, while temporal snapshots $T(t)$ capture long-range evolution. *Third, a graph adaptation and attention block* fuses the sequential and temporal snapshots and encodes them with relation-aware GATv2Conv layers. Adaptive edge weighting, which integrates agreement-aware similarity, temporal decay, and gating, and a lightweight *Feedback Mechanism*, further refines the node representations without full recomputation. *Fourth, a Causal Attention and Fusion mechanism* then aggregates outputs from different graph paths, guided by a *Causal Alignment Loss* to preserve chronological order and reduce temporal leakage. *Finally, node-level logits are aggregated via mean pooling and projected through an MLP-softmax classifier to produce event-level veracity predictions* (True, False, Unverified, Non-rumor). The following subsections elaborate on each block and its role in the end-to-end DCAGTN pipeline.

3.2. Node representations and feature construction

In DCAGTN, each node in the dynamic interaction graph represents a social media post, either a source message or a reply, augmented with both *semantic* and *structural* features, mirroring real-world information diffusion patterns. This dual-modality feature design allows the model to capture complementary information: the content features of what is being said and the interaction dynamics of how information spreads. This design is motivated by recent literature demonstrating that both content and propagation context are critical for rumor detection accuracy [13,25,26,35]. Our implementation integrates a RoBERTa-based semantic encoder with explicit relational and temporal interaction features, as described below.

1. Text Feature Extraction. For each post i , we compute a contextual semantic embedding $\mathbf{z}_i^T \in \mathbb{R}^{d_T}$ using the [CLS] output of a pretrained RoBERTa-base model:

$$\mathbf{H}_i^T = \text{RoBERTa}(T_i), \quad \mathbf{z}_i^T = \mathbf{H}_i^T[0] \in \mathbb{R}^{768} \quad (2)$$

where T_i is the tokenized text of post i and $d_T = 768$. This approach leverages transformer-based contextual encoding, capturing rich linguistic semantics beyond what static word2vec or LSTM models achieve [45,46].

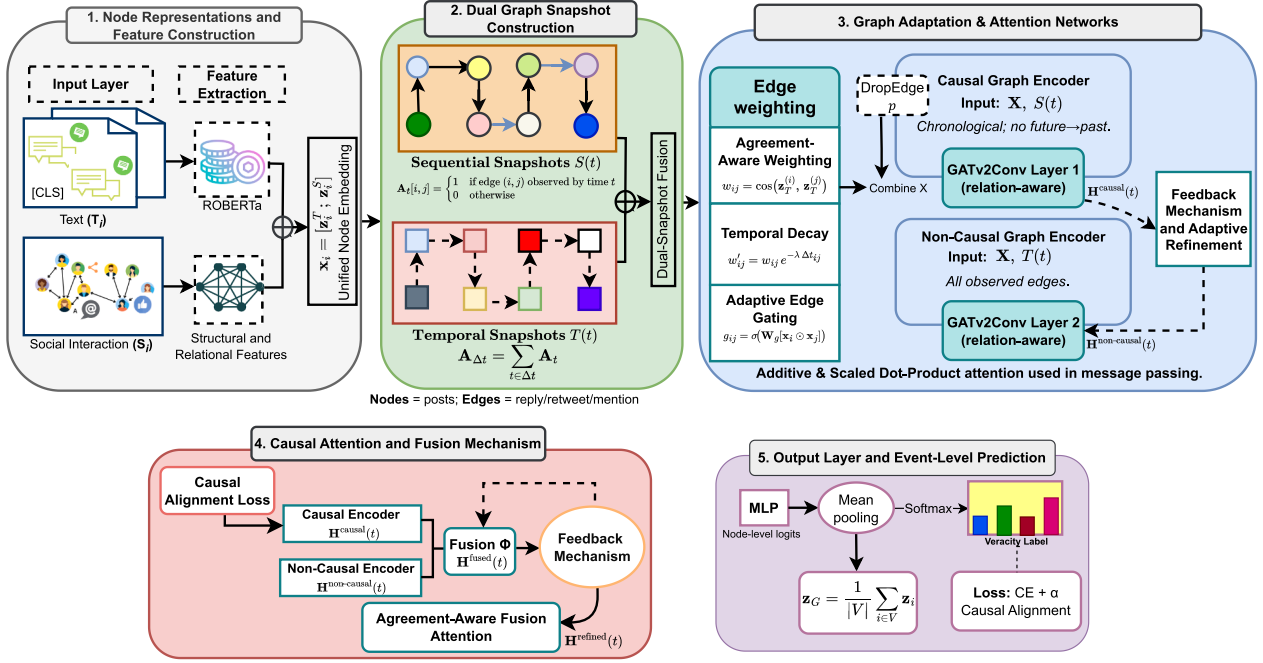


Fig. 1. Proposed DCAGTN model: RoBERTa-based node features from sequential $S(t)$ and temporal $T(t)$ snapshots pass through dual relation-aware GATv2Conv encoders, fused via causal attention with agreement-aware weighting, refined by feedback, and regularized by causal alignment loss for real-time rumor detection.

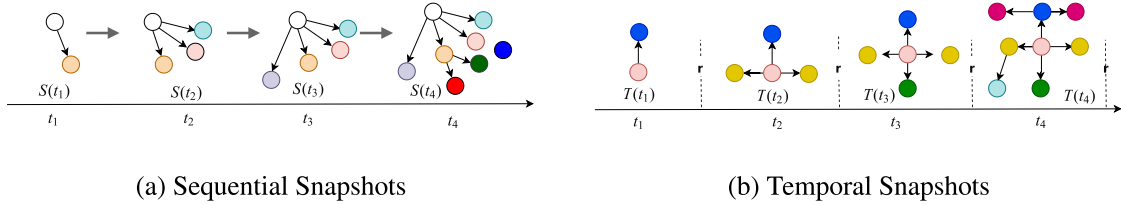


Fig. 2. Graph Snapshots: (a) Sequential snapshots $S(t)$ capture the real-time progression of conversation threads, preserving chronological interaction flow. (b) Temporal snapshots $T(t)$ aggregate interactions within fixed time windows, offering a coarse-grained view of network states over time.

2. Structural and Relational Features. Each post i is further characterized by a structural embedding $\mathbf{z}_i^S \in \mathbb{R}^{d_S}$ reflecting its interaction-based context within the propagation graph:

- **Edge type** $r_{ij} \in \{0, 1, 2\}$ (e.g., reply, retweet, mention), encoded as a one-hot vector.
- **Timestamp** $t_{ij} \in \mathbb{R}_+$, preprocessed via min-max normalization.

For every edge (i, j) , we construct the feature vector:

$$\mathbf{f}_{ij} = [\text{OneHot}(r_{ij}); \text{Norm}(t_{ij})] \quad (3)$$

To obtain the node's overall structural embedding, we aggregate across all neighbors $\mathcal{N}(i)$:

$$\mathbf{z}_i^S = \text{AGG}(\{\mathbf{f}_{ij} \mid j \in \mathcal{N}(i)\}) \quad (4)$$

where AGG denotes mean pooling over the feature vectors of neighboring nodes [47].

3. Unified Node Embedding. The final node feature is obtained by concatenating the semantic and structural representations:

$$\mathbf{x}_i = [\mathbf{z}_i^T; \mathbf{z}_i^S] \in \mathbb{R}^{d_x}, \quad d_x = d_T + d_S \quad (5)$$

This joint vector $\mathbf{x}_i$ is then propagated through the graph encoder modules. In contrast with models such as BiGCN [13], which process textual and structural information in isolation, or AGAD [21], which relies on static GAT attention with adversarial contrastive learning, or FGDGNN [25], which captures snapshot-level temporal transitions, and PSGT [26], which applies structure-aware transformers, our framework enables early integration by injecting both semantic (RoBERTa)

and structural (typed, timestamped) features directly at the node level. Hence, we preserve temporal and relation-type features throughout the network, facilitating dual-snapshot construction and supporting causal attention.

3.3. Dual graph snapshot construction

Rumor propagation on social media is inherently dynamic, evolving through sequences of replies, retweets, and interactions that unfold over time [13,16]. Static graphs cannot capture this temporal evolution, often resulting in label leakage and suboptimal early detection [35]. To address this, we employ a dual-snapshot mechanism to explicitly model both short-term and long-term information diffusion as seen in Fig. 2.

1. Sequential Snapshots $S(t)$. At each time step t , we define a sequential snapshot graph $S(t) = (V_t, E_t)$, where V_t is the set of posts active up to t , and E_t the set of observed interactions. The adjacency matrix A_t for $S(t)$ is:

$$A_t[i, j] = \begin{cases} 1 & \text{if edge } (i, j) \text{ observed by time } t \\ 0 & \text{otherwise.} \end{cases} \quad (6)$$

This construction respects chronological order, preventing information from future interactions from leaking into earlier predictions. Sequential snapshots $(S(t))$ are cumulative, preserving strict chronological order and capturing fine-grained causal propagation across time.

2. Temporal Snapshots $T(t)$. To alleviate the sparsity of early snapshots and capture longer-range dependencies, we aggregate interactions

over coarser time intervals Δt , producing temporal snapshots $T(t) = (V_{\Delta t}, E_{\Delta t})$:

$$A_{\Delta t} = \sum_{t \in \Delta t} A_t \quad (7)$$

This sum operation creates denser graphs for each period, enabling the model to identify global patterns that may only emerge later in the event. Temporal snapshots $T(t)$, by contrast, are segmented into fixed time windows to capture coarser, macroscopic diffusion patterns. This dual design allows the model to maintain causal order while also summarizing broader temporal context.

3. Dual-Snapshot Fusion. Our model leverages both sequential and temporal snapshots during training, enabling a joint learning process:

- Sequential snapshots $(S(t))$ capture short-term, fine-grained diffusion essential for rapid rumor detection.
- Temporal snapshots $(T(t))$ encode macro-scale regularities and long-range dependencies, mitigating the instability of sparse early data.

For each event, the node and edge sets are maintained across snapshots to ensure consistent feature propagation. This strategy addresses the limitations of static models such as DeepWalk [7] and earlier dynamic GCNs [16], which do not jointly optimize across timescales.

To further prevent leakage, event-level k-fold cross-validation is enforced, ensuring that all snapshots from the same conversation remain within a single fold and no temporal information leaks across training and validation.

4. Implementation Details. Our snapshots are generated via:

$$\text{Snap}(G, \mathcal{T}, N_{\text{snap}}) = \{G_{S(t_1)}, \dots, G_{S(t_n)}, G_{T(\Delta t_1)}, \dots, G_{T(\Delta t_m)}\}, \quad (8)$$

where N_{snap} is a hyperparameter for the number of snapshots. Snapshots are serialized and fed to the model, supporting efficient mini-batch and event-level k-fold cross-validation. Unlike methods that use only static or single-window temporal graphs, our approach aligns with best practices in recent dynamic GNN literature [17–19] by maintaining both real-time responsiveness and long-range trend awareness.

3.4. Graph adaptation and attention networks

After constructing the dual snapshots, DCAGTN must propagate and adapt node representations to reflect evolving multi-relational, multi-scale interaction patterns. To this end, we develop a *Graph Adaptation Network* that integrates relation-aware convolution, adaptive edge gating, and attention mechanisms.

1. Relation-Aware Graph Convolutions. To capture heterogeneous interactions (e.g., reply, retweet, mention), we model the event as a multi-relational graph $G = (V, E, R)$, where R denotes relation types. Each node's features are updated using a stack of GATv2Conv layers [48], which extend the expressivity of classical RGCN [47]:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{r \in R} \sum_{j \in \mathcal{N}_r(i)} \alpha_{ij}^r \mathbf{W}_r \mathbf{h}_j^{(l)} \right) \quad (9)$$

where $\mathcal{N}_r(i)$ denotes neighbors via relation r , $\mathbf{W}_r$ is a relation-specific weight matrix, and α_{ij}^r is an attention coefficient computed for each relation.

2. Adaptive Edge Gating and Agreement-Aware Weighting. Unlike previous works using static or uniform edge weights [13,21], we introduce an *edge gating mechanism* and agreement-aware weighting, which modulates the strength of each edge dynamically:

$$g_{ij} = \text{Sigmoid}(\mathbf{W}_g [\mathbf{x}_i \odot \mathbf{x}_j]) \quad (10)$$

where $\odot$ is element-wise multiplication and $\mathbf{W}_g$ are learnable weights. If semantic agreement between parent and child posts (computed via cosine similarity of [CLS] embeddings) is high, the edge is upweighted; conversely, older interactions are downweighted using a temporal decay factor

$$w_{ij}^{\text{temp}} = \exp(-\lambda \cdot (t_{\text{obs}} - t_{ij})) \quad (11)$$

where t_{ij} is the timestamp of the interaction and $\lambda > 0$ controls the rate of decay. This combination favours edges that are both semantically consistent and temporally relevant, strengthening the influence of recent, rumor-consistent propagation paths.

3. Dual Graph Encoders (Causal & Non-Causal). In our model design, two independent GraphEncoder modules are instantiated:

$$\mathbf{H}^{\text{causal}} = \text{GraphEnc}_c(\mathbf{X}, \mathbf{A}_c, \mathbf{R}, \mathbf{w}) \quad (12)$$

$$\mathbf{H}^{\text{non-causal}} = \text{GraphEnc}_{nc}(\mathbf{X}, \mathbf{A}_{nc}, \mathbf{R}, \mathbf{w})$$

The causal path strictly follows event chronology; the non-causal path allows all observed edges, capturing broader propagation motifs.

4. Multi-Head Attention and Readout. To further enhance contextual relevance, we employ both additive and scaled dot-product attention over node pairs, inspired by [22,49]. For additive attention:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}_h \mathbf{i} \parallel \mathbf{W}_h \mathbf{j}]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}_h \mathbf{i} \parallel \mathbf{W}_h \mathbf{k}]))} \quad (13)$$

For scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^\top}{\sqrt{d_k}} \right) V \quad (14)$$

where Q, K, V are query, key, value matrices from node representations.

5. DropEdge Regularization. To prevent overfitting, we apply DropEdge [50], randomly removing a fraction p of edges during training:

$$\mathbf{A}' = \mathbf{A} \odot \mathbf{M} \quad (15)$$

where $\mathbf{M}$ is a Bernoulli mask. This compels the network to generalize across diverse substructures, as realized in both the training and data loading pipelines.

6. Feedback Mechanism and Adaptive Refinement. Distinct from classical GNN pipelines, DCAGTN integrates a lightweight feedback mechanism that refines node embeddings by incorporating prior states, enabling rapid incremental updates instead of full graph recomputation [11]. Rather than recalculating all node representations whenever new propagation edges appear, the feedback module adaptively fuses the previous and current node information under a learned gating process. The detailed mathematical formulation of this gating vector and refinement rule is presented in Section 3.5 (Eqs. 21–22), where we show how the gate $G(t)$ dynamically balances the influence of historical and newly fused features based on temporal context. This design allows DCAGTN to efficiently adapt to evolving interactions in real time while maintaining chronological consistency and preserving the integrity of causal information flow.

3.5. Causal attention and fusion mechanism

A central innovation of our DCAGTN is the introduction of a **dual-path causal attention framework** that explicitly disentangles temporal and structural learning. Our approach leverages two parallel graph encoders to process causal (chronological) and non-causal (full-graph) propagation. The resulting representations are adaptively fused and then iteratively refined via a feedback mechanism. In DCAGTN, the term *causal* refers to the strict preservation of chronological order within dynamic propagation rather than counterfactual or do-calculus reasoning. This design enforces one-way temporal consistency across snapshots, aligning feature updates with event chronology to prevent future-to-past information leakage.

1. Causal and Non-Causal Graph Encoders. Given the dual snapshots, we maintain two distinct node embedding flows:

$$\mathbf{H}^{\text{causal}}(t) = \text{GraphEnc}_c(\mathbf{X}, \mathbf{A}_c(t), \mathbf{R}, \mathbf{W}^c) \quad (16)$$

$$\mathbf{H}^{\text{noncausal}}(t) = \text{GraphEnc}_{nc}(\mathbf{X}, \mathbf{A}_{nc}(t), \mathbf{R}, \mathbf{W}^{nc}) \quad (17)$$

where $\mathbf{X}$ denotes node features, $\mathbf{A}_{\text{causal}}$ encodes temporally valid edges (no future-to-past leakage), and $\mathbf{A}_{\text{noncausal}}$ allows any observed interaction. These parallel encoders ensure that the model respects real-world event order while also benefiting from broader propagation motifs.

2. *Causal Attention Fusion Layer.* To exploit both temporal (causal) and global (non-causal) patterns, we fuse the two streams using a learnable layer that serves as a fusion block:

$$\mathbf{H}^{\text{fused}}(t) = \Phi(\mathbf{H}^{\text{causal}}(t), \mathbf{H}^{\text{noncausal}}(t)) \quad (18)$$

where Φ denotes a fusion operator (e.g., concatenation followed by linear projection and nonlinearity).

3. *Agreement-Aware Fusion Attention.* DCAGTN leverages semantic agreement between post pairs as edge weights. For each edge (i, j) , the agreement score w_{ij} is computed as the cosine similarity of their [CLS] embeddings:

$$w_{ij} = \cos(\mathbf{z}_i^{(i)}, \mathbf{z}_j^{(j)}) \quad (19)$$

These weights directly modulate message-passing strength, enabling the model to emphasize high-agreement (rumor-consistent) threads while suppressing noisy or contradictory ones. This mechanism is operationalized through agreement-aware edge weighting during both snapshot construction and the model forward pass.

4. *Temporal Decay and Edge Gating.* Temporal relevance is enforced via a learnable decay:

$$w'_{ij} = w_{ij} \cdot \exp(-\lambda \cdot \Delta t_{ij}) \quad (20)$$

where Δt_{ij} is the normalized time difference between parent and child, and λ is a learnable decay parameter. This approach, realized in both data preprocessing and the model, ensures that distant, outdated interactions have proportionally reduced influence, an explicit improvement over static attention or unweighted propagation in prior GNNs [13,16]. Implementation-wise, agreement-aware similarity and temporal decay scores are precomputed during preprocessing and stored with each snapshot, while the DCAGTN forward pass refines these weights through a learnable decay parameter and gating. This ensures consistency between data preparation and model training without introducing temporal leakage.

5. *Feedback Mechanism.* Let $\mathbf{H}^{\text{fused}}(t) \in \mathbb{R}^{n_t \times d}$ be node embeddings after dual-path fusion at snapshot t , and let $\tilde{\mathbf{H}}(t-1)$ denote the detached (stop-gradient) refined states from the previous snapshot, aligned to the currently active node set $\mathcal{V}_t$ (missing nodes are zero-filled). We compute a per-node gate

$$\mathbf{G}(t) = \sigma(\mathbf{W}_g [\mathbf{H}^{\text{fused}}(t) \parallel \tilde{\mathbf{H}}(t-1) \parallel \Delta \mathbf{t}(t)] + \mathbf{b}_g), \quad (21)$$

where $[\cdot \parallel \cdot]$ is concatenation, $\Delta \mathbf{t}(t) \in \mathbb{R}^{n_t \times 1}$ stores normalized inter-snapshot time gaps, σ is the sigmoid, and $\mathbf{W}_g \in \mathbb{R}^{d \times (2d+1)}$, $\mathbf{b}_g \in \mathbb{R}^d$. The gated refinement is

$$\mathbf{H}^{\text{refined}}(t) = \mathbf{G}(t) \odot \phi(\mathbf{W}_u \mathbf{H}^{\text{fused}}(t)) + (1 - \mathbf{G}(t)) \odot \tilde{\mathbf{H}}(t-1), \quad (22)$$

with $\phi = \text{ReLU}$ and $\mathbf{W}_u \in \mathbb{R}^{d \times d}$. We perform one refinement step per snapshot (additional inner iterations did not help in validation), which costs $\mathcal{O}(n_t d^2)$ and avoids full graph recomputation.

6. *Causal Alignment Loss (targets and optimization).* To enforce directional (past→future) consistency, we align the causal stream at time t to a temporally valid target $\mathbf{y}_t$ built only from past information:

$$\mathbf{y}_t = \mathbf{M}_t \odot \text{SG}(\mathbf{H}^{\text{causal}}(t-1)) + (1 - \mathbf{M}_t) \odot \text{SG}(\mathbf{W}_p \mathbf{H}^{\text{fused}}(t)), \quad (23)$$

where $\text{SG}(\cdot)$ is stop-gradient, $\mathbf{M}_t \in \{0, 1\}^{n_t \times 1}$ masks nodes already present at $t-1$, and $\mathbf{W}_p \in \mathbb{R}^{d \times d}$ provides a warm start for new nodes. The alignment objective is

$$\mathcal{L}_{\text{causal}} = \frac{1}{\sum_t n_t} \sum_{t=1}^T \left\| \mathbf{H}^{\text{causal}}(t) - \mathbf{y}_t \right\|_F^2, \quad (24)$$

optionally reweighted by time gaps via $\omega_t = \exp(-\lambda \Delta t_t)$ when larger inter-snapshot intervals should be down-weighted. This directly supervises continuous representations, is numerically stable on sparse early snapshots, and never uses future states to supervise the past.

Training details. We optimize $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{causal}}$ with a linear warm-up of α from 0 to α_{max} over the first E_{warm} epochs, gradient clipping at 1.0, early stopping on validation macro-F1, and strict detachment

of all targets in $\mathbf{y}_t$ to prevent collapse. Snapshot mini-batches are independent to preclude future-to-past leakage.

3.6. Output layer and event-level prediction

After adaptive causal refinement, the DCAGTN architecture performs event-level rumor prediction by aggregating node-level logits and mapping them to veracity classes via a dedicated output module. This stage is critical, as it ensures the model provides an interpretable and contextually grounded decision for each rumor event.

1. *Node-Level Logit Generation.* The final, refined node representations $\mathbf{H}^{\text{refined}} \in \mathbb{R}^{N \times d}$ (where N is the number of nodes and d the feature dimension) are projected into class logits using a multi-layer perceptron:

$$\mathbf{z}_i = \text{MLP}(\mathbf{h}_i^{\text{refined}}), \quad i = 1, \dots, N \quad (25)$$

where $\text{MLP}(\cdot)$ comprises one or more fully connected layers with non-linear activation functions.

2. *Mean-Pooling Graph Readout.* To transform node-level predictions into a single event-level decision, we aggregate node logits using mean pooling:

$$\mathbf{z}_G = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i \quad (26)$$

This approach preserves the influence of all posts, enabling robust, context-aware prediction, an improvement over earlier rumor detectors that relied on either root-only classification [35] or simplistic pooling.

3. *Classification and Softmax.* The pooled graph vector $\mathbf{z}_G$ is passed through a final MLP and softmax to yield class probabilities:

$$\mathbf{H}' = \phi(\mathbf{W}_h \mathbf{z}_G + \mathbf{b}_h) \quad (27)$$

$$\mathbf{P} = \text{Softmax}(\mathbf{W}_o \mathbf{H}' + \mathbf{b}_o) \quad (28)$$

where $\mathbf{W}_h$ and $\mathbf{W}_o$ are trainable weight matrices, $\mathbf{b}_h$ and $\mathbf{b}_o$ are biases, and ϕ is the ReLU activation.

The predicted class label for the event is given by:

$$\hat{L} = \arg \max_j P_j, \quad j \in \{\text{NR}, \text{FR}, \text{TR}, \text{UR}\} \quad (29)$$

where NR: Non-Rumor, FR: False Rumor, TR: True Rumor, UR: Unverified Rumor.

4. *Loss Function and Optimization.* The overall training objective combines the event-level cross-entropy loss and the causal alignment regularizer:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{causal}}, \quad (30)$$

where $\mathcal{L}_{\text{CE}}$ ensures correct veracity prediction and $\mathcal{L}_{\text{causal}}$ enforces chronological consistency between snapshots. Training follows an event-level k-fold cross-validation setup to avoid temporal leakage, with early stopping and best-checkpoint selection for stability.

3.7. Reproducibility Statement

The complete procedures of our model are presented in [Algorithm 1](#), while [Appendix A](#) provides detailed pseudocode and implementation information for all components of DCAGTN, including data preprocessing, dual-snapshot encoding, causal attention fusion, feedback refinement, and loss optimization.

4. Experiments and evaluation

We present the datasets, baseline methods, experimental settings, and analyze model performance.

Algorithm 1 DCAGTN algorithm description.

Input: Node set V , Edge set E , Raw texts $\{T_i\}$, Graph attributes (types, times), Snapshots $\{S(t), T(t)\}_{t=1}^T$, Parameters Θ

Output: Event-level veracity label $\hat{Y}$

- 1: **Preprocessing:**
 Encode texts via RoBERTa to get $\mathbf{z}_T^i$; aggregate structural features into $\mathbf{z}_S^i$; form node feature $\mathbf{x}_i = [\mathbf{z}_T^i; \mathbf{z}_S^i]$.
 Compute agreement-aware similarities and temporal decay weights for edges; store per snapshot (no future-to-past usage).
- 2: **Dual-Snapshot Construction:**
 Build sequential $S(t)$ (chronology-preserving) and temporal $T(t)$ (coarser aggregation) graphs; enforce event-level fold separation.
- 3: **for** $t = 1$ **to** T **do**
- 4: **Graph Encoding (two streams):**
 $\mathbf{H}^{\text{causal}}(t) \leftarrow \text{GraphEnc}_c(\mathbf{X}, \mathbf{A}_c(t), \mathbf{R}, w)$
 $\mathbf{H}^{\text{noncausal}}(t) \leftarrow \text{GraphEnc}_{nc}(\mathbf{X}, \mathbf{A}_{nc}(t), \mathbf{R}, w)$
- 5: **Fusion:**
 $\mathbf{H}^{\text{fused}}(t) \leftarrow \Phi(\mathbf{H}^{\text{causal}}(t), \mathbf{H}^{\text{noncausal}}(t))$
- 6: **Feedback Refinement (one step):**
 Align previous refined states to current nodes: $\tilde{\mathbf{H}}(t-1)$ (missing nodes zero-filled; stop-gradient).
 $\mathbf{G}(t) \leftarrow \sigma(\mathbf{W}_g [\mathbf{H}^{\text{fused}}(t) \parallel \tilde{\mathbf{H}}(t-1) \parallel \Delta t(t)] + \mathbf{b}_g)$
 $\mathbf{H}^{\text{refined}}(t) \leftarrow \mathbf{G}(t) \odot \phi(\mathbf{W}_u \mathbf{H}^{\text{fused}}(t)) + (\mathbf{1} - \mathbf{G}(t)) \odot \tilde{\mathbf{H}}(t-1)$
- 7: **Causal Alignment Target:**
 Build mask $\mathbf{M}_t$ for nodes present at $t-1$.
 $\mathbf{y}_t \leftarrow \mathbf{M}_t \odot \text{SG}(\mathbf{H}^{\text{causal}}(t-1)) + (\mathbf{1} - \mathbf{M}_t) \odot \text{SG}(\mathbf{W}_p \mathbf{H}^{\text{fused}}(t))$
- 8: **Losses at snapshot t :**
 $\mathcal{L}_{\text{causal}}(t) \leftarrow \|\mathbf{H}^{\text{causal}}(t) - \mathbf{y}_t\|_F^2$
 (Optionally weight by $\omega_t = \exp(-\lambda \Delta t_t)$.)
- 9: **end for**
- 10: **Event-level Prediction:**
 Obtain node logits $\mathbf{z}_i \leftarrow \text{MLP}(\mathbf{h}_i^{\text{refined}}(t))$ from the latest snapshot t .
 Aggregate $\mathbf{z}_G = \frac{1}{N} \sum_i \mathbf{z}_i$; compute $\mathbf{P} = \text{Softmax}(\text{MLP}(\mathbf{z}_G))$; output $\hat{Y} = \arg \max_j P_j$.
- 11: **Training Objective and Protocol:**
 $\mathcal{L}_{\text{CE}}$ at event level; $\mathcal{L}_{\text{causal}} = \frac{1}{\sum_i n_i} \sum_{t=1}^T \mathcal{L}_{\text{causal}}(t)$.
 $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \alpha \mathcal{L}_{\text{causal}}$; linearly warm-up α to α_{max} over E_{warm} .
 Train with Adam; gradient clipping (1.0); early stopping on validation macro-F1; event-level k-fold CV; detach all targets in $\mathbf{y}_t$.

4.1. Datasets

We used three publicly available rumor detection datasets, Twitter15, Twitter16, and Weibo, to evaluate the performance of our model. The Twitter15 and Twitter16 datasets include four veracity labels representing Non-Rumor, False Rumor, True Rumor, and Unverified Rumor. In contrast, the Weibo dataset categorizes posts into two veracity labels, indicating False Rumor and True Rumor. Table 1 summarizes key statistics for each dataset, including the number of source posts, false rumors, true rumors, non-rumors, unverified rumors, and the overall number of posts and users.

4.2. Baseline methods

We compare DCAGTN with key baselines, including traditional and deep learning approaches. We selected them based on their relevance and methodological contributions in rumor detection.

- RvNN [35]: Enhanced tree structures of SVM-TK [34] with tree-structured recursive neural networks (RvNNs) to capture the propagation characteristics of rumors. This method is included for its recursive neural network approach to improve rumor propagation modelling.

Table 1

Datasets information.

Statistic	Twitter15	Twitter16	Weibo
# of source posts	1490	818	4664
# of false rumors	370	205	2313
# of true rumors	372	205	–
# of non-rumors	374	205	2351
# of unverified rumors	374	203	–
# of posts	331,612	204,820	3,805,656
# of users	276,663	173,487	2,746,818
Avg. max time /event	1337 Hours	848 Hours	2461 Hours
Avg. # of posts / event	222.6	250.4	816.0
Max. # of posts / event	1768	2765	59,318

- BiGCN [13]: Applied a bidirectional graph convolutional network to capture both upward and downward propagation features. This model was chosen for its bidirectional GCN approach, which enhances feature capture from propagation paths.
- D_{YN}GCN [16]: Captured the evolving pattern of rumor propagation by aggregating sequential and temporal snapshots, showing improvements over static methods. It is selected for its dynamic approach to handling temporal data in rumor detection.
- DA-GCN [36]: Expanded on GCN integration with a dual attention mechanism, improving early rumor detection accuracy by filtering false interactions. This method is included because of its innovative dual attention mechanism that improves detection accuracy.
- DDGCN [18]: Integrated dual-dynamic graph convolutional networks and graph-aware mechanisms to provide an early, fine-grained classification of rumors. It is selected for its fine-grained classification capabilities and dynamic graph approach.
- CDGTN [19]: Introduced the CDGTN model, advancing the field towards more predictive and generalizable models by strategically handling unseen data representations. This model was chosen for its causal and masked transformer approach, which improves the generalizability of our model and prediction accuracy.
- AGAD [21]: Introduced a GAT-based architecture enhanced with attention filtering, adversarial training, and dual contrastive learning to improve rumor detection under noisy and adversarial conditions. This model was chosen for its robust integration of structural denoising and contrastive representation learning.
- PSQT [26]: Developed a propagation structure-aware graph transformer that jointly models structural and temporal features using attention over propagation paths. This model was selected for its robust transformer-based modeling and interpretability in dynamic social media environments.
- FGDGNN [25]: Proposed a fine-grained dynamic graph neural network that captures intra- and inter-snapshot dependencies through temporal snapshot construction and embedding refinement. This method was included for its emphasis on temporal coherence and early-stage rumor detection under noisy conditions.

4.3. Implementation details

We implemented our DCAGTN framework in PyTorch 2.3 with Torch Geometric 2.5 and Hugging Face Transformers (v4.40) for deep learning and text encoding. Textual features were extracted using a pretrained RoBERTa-base encoder (hidden size 768), which remained frozen during all experiments. Social interaction features were embedded using a two-layer GraphEncoder comprising multi-head GATv2Conv layers (hidden size 256 per head, 8 heads). These layers explicitly modeled relation types and incorporated both agreement-aware and temporal decay edge weighting. We applied DropEdge [50] regularization with a rate of 0.2 to promote robustness through stochastic edge masking. Textual and structural embeddings were concatenated into unified node vectors of dimension 1024.

We optimized our model using the Adam optimizer (learning rate 1×10^{-4} , weight decay 1×10^{-4}) with a batch size of 4 and cross-entropy loss. Each experiment ran for up to 30 epochs per dataset, with early stopping based on the validation macro F1-score. We employed stratified 5-fold cross-validation to maintain balanced class distributions. All experiments were conducted on a workstation equipped with a multi-core CPU and a modern CUDA-compatible GPU (with at least 24 GB of VRAM).

4.4. Computational complexity and scalability

We summarize the time and memory costs of DCAGTN and how the dual snapshots and feedback refinement affect scalability. Let $n=|V|$ and $m=|E|$ denote the number of nodes and edges per snapshot, L the number of GATv2Conv layers, H the number of attention heads, d the hidden size, $|R|$ the number of relation types, N_{snap} the number of snapshots used per event (sequential + temporal), and p the DropEdge rate (so the expected active edges are $m' = (1-p)m$). Let d_u denote the projection width used in feedback updates (with $d_u \ll d$ in our implementation).

(1) *Text encoding (frozen)*. RoBERTa is used only in forward mode. If N_{tok} is the total number of tokens across posts in the batch, the one-time cost is $\mathcal{O}(N_{\text{tok}})$ with no backprop. Since embeddings are cached and reused across snapshots, this cost does not scale with N_{snap} during training.

(2) *Snapshot construction*. Building sequential $S(t)$ and temporal $T(t)$ snapshots requires linear passes over interactions: $\mathcal{O}(N_{\text{snap}}(n+m))$. Temporal windows use a sum over per-step adjacencies (Eq. (11)), which is also linear.

(3) *Relation-aware GATv2Conv (per snapshot)*. For a single snapshot, one GATv2Conv layer with H heads performs (i) linear projections on nodes and (ii) attention/message passing on edges:

$$\mathcal{O}(H n d^2) + \mathcal{O}(H m' d) = \mathcal{O}(H(n d^2 + m' d)). \quad (31)$$

With L layers, the per-snapshot time is $\mathcal{O}(L H(n d^2 + m' d))$. Relation-specific parameters introduce constant factors but do not change the order (edges are partitioned across $r \in R$ with $\sum_r m'_r = m'$). Memory per snapshot is $\mathcal{O}(n d + H m')$ for node states and attention coefficients.

(4) *Dual-snapshot encoders and fusion*. DCAGTN processes the causal and non-causal streams independently and fuses them:

$$\underbrace{2 \mathcal{O}(L H(n d^2 + m' d))}_{\text{two encoders}} + \underbrace{\mathcal{O}(n d^2)}_{\text{fusion projection}} \quad (32)$$

$$= \mathcal{O}(L H(n d^2 + m' d)).$$

where the fusion cost is lower order in practice. Over N_{snap} snapshots in a forward pass, the time is

$$\mathcal{O}(N_{\text{snap}} L H(n d^2 + m' d)). \quad (33)$$

Backward propagation has the same order.

Peak memory is dominated by activations and attention maps:

$$\mathcal{O}(N_{\text{snap}}(n d + H m')) \quad (34)$$

for the two paths combined. To keep this within GPU limits, we mini-batch over snapshots, which bounds runtime and memory in practice.

(5) *Agreement-aware gating and temporal decay*. Computing agreement scores and decay weights is linear in edges and done once per snapshot (cached): $\mathcal{O}(m')$ time and $\mathcal{O}(m')$ memory. During training, these weights are read and used as multiplicative factors with negligible extra cost beyond message passing.

(6) *Feedback refinement*. The gated refinement applies a lightweight projection to current node states:

$$\mathcal{O}(n d d_u) + \mathcal{O}(n d) = \mathcal{O}(n d d_u), \quad d_u \ll d. \quad (35)$$

Table 2

Rumor detection results on weibo dataset.

Model	Acc.	FR			TR		
		Prec.	Rec.	F1	Prec.	Rec.	F1
RvNN	0.908	0.912	0.897	0.905	0.904	0.918	0.911
BiGCN	0.961	0.961	0.964	0.961	0.962	0.962	0.960
D _{YN} GCN	0.936	0.930	0.942	0.936	0.935	0.938	0.936
DA-GCN	0.944	0.941	0.946	0.944	0.947	0.941	0.944
DDGCN	0.948	0.953	0.948	0.950	0.951	0.946	0.948
CDGTN	0.965	0.966	0.966	0.966	0.965	0.968	0.966
AGAD	0.967	0.970	0.968	0.969	0.968	0.967	0.968
PSGT	0.924	0.917	0.930	0.923	0.932	0.918	0.924
FGDGNN	0.941	0.935	0.945	0.940	0.947	0.936	0.941
DCAGTN*	0.969	0.971	0.967	0.970	0.969	0.963	0.971

In streaming updates, only the currently active nodes n_i (and their incident edges m_i) are touched, giving $\mathcal{O}(H m_i d + n_i d d_u)$ per micro-update rather than recomputing over all n, m .

(7) *Causal-consistency alignment loss*. The MSE term supervises causal-path representations over snapshots:

$$\mathcal{O}(N_{\text{snap}} n d) \quad (36)$$

with negligible memory overhead beyond the stored states.

(8) *Output and readout*. Node-level MLPs and mean pooling incur $\mathcal{O}(n d^2)$ per snapshot and $\mathcal{O}(n d)$ for pooling, both dominated by graph propagation.

A. *Per-epoch cost*. Let B be the number of events per batch and κ the average number of snapshots per event used in a step ($\kappa \leq N_{\text{snap}}$ due to snapshot mini-batching). The dominant per-step training cost is

$$\boxed{\mathcal{O}(B \kappa L H(n d^2 + m' d))} \quad (37)$$

with memory $\mathcal{O}(B \kappa(n d + H m'))$. Because m' typically dominates n in social graphs and DropEdge reduces m' by $(1-p)$, the practical scaling is linear in the number of active edges per step. Dual snapshots and two encoders introduce constant factors (≈ 2) while preserving linear scaling.

B. *Inference and early-stage operation*. In early windows, both n and m' are small, so inference remains efficient:

$$\mathcal{O}(\kappa L H(n d^2 + m' d)), \quad (38)$$

and the feedback update allows incremental assimilation of new interactions with $\mathcal{O}(H m_i d + n_i d d_u)$ cost, avoiding full re-computation.

C. *Empirical footprint*. On a single RTX 3090 (24 GB), our configuration ($L=2$, $H=8$, frozen RoBERTa, $d=256$ per head) fits comfortably; peak memory is consistent with the $\mathcal{O}(n d + H m')$ activation profile, and per-batch latency follows the linear edge scaling predicted above. This aligns with our observed stable throughput across Twitter15/16 and Weibo.

4.5. Results analysis

Tables 2 and 3 present the comparative performance of DCAGTN against recent state-of-the-art models. Several key findings can be drawn from these results:

(1) *Superior Accuracy and Robustness Across Datasets*. DCAGTN consistently achieves the highest accuracy across all datasets, surpassing strong baselines such as AGAD, CDGTN, and FGDGNN. This highlights the effectiveness of our dual-snapshot architecture, which jointly captures short-term interactions and long-range temporal propagation. Notably, on the Twitter16 dataset, DCAGTN improves upon AGAD by 0.7% in accuracy and 0.5% in FR-F1, highlighting its robustness in capturing evolving rumor dynamics.

Table 3
Rumor detection results on Twitter15 and Twitter16 datasets.

Model	Twitter 15 (F1)					Twitter 16 (F1)				
	Accuracy	NR	FR	TR	UR	Accuracy	NR	FR	TR	UR
RvNN	0.723	0.682	0.758	0.821	0.654	0.737	0.662	0.743	0.835	0.708
BiGCN	0.886	0.891	0.860	0.930	0.864	0.880	0.847	0.869	0.937	0.865
D _{YN} GCN	0.827	0.837	0.769	0.820	0.746	0.836	0.880	0.804	0.853	0.741
DA-GCN	0.905	0.959	0.895	0.914	0.852	0.902	0.894	0.872	0.928	0.913
DDGCN	0.908	0.958	0.851	0.903	0.876	0.910	0.920	0.869	0.951	0.907
CDGTN	0.916	0.947	0.951	0.912	0.859	0.929	0.874	0.949	0.956	0.931
AGAD	0.935	0.893	0.984	0.947	0.913	0.950	0.926	0.952	0.993	0.915
PSGT	0.808	0.804	0.827	0.845	0.760	0.819	0.798	0.835	0.850	0.760
FGDGNN	0.840	0.845	0.860	0.870	0.832	0.863	0.850	0.870	0.880	0.830
DCAGTN*	0.939	0.963	0.985	0.949	0.918	0.957	0.932	0.957	0.971	0.939

Table 4
Unique features of DCAGTN and key baseline models (evaluated on Twitter15 and Twitter16).

Model	Causal Feedback	Alignment Mechanism	Dual Snapshots	UR (F1) ₁₅	Acc. ₁₅	UR (F1) ₁₆	Acc. ₁₆
DA-GCN	✗	✗	✗	0.852	0.905	0.913	0.902
DDGCN	✗	✗	✗	0.876	0.908	0.907	0.910
CDGTN	✓	✗	✗	0.859	0.916	0.931	0.929
AGAD	✗	✗	✗	0.913	0.935	0.915	0.950
DCAGTN*	✓	✓	✓	0.918	0.939	0.939	0.957

(2) *Advantage of Causal Attention and Alignment.* Compared to models that focus solely on structural or temporal modeling, DCAGTN integrates causal attention and causal alignment loss to preserve event chronology and mitigate temporal leakage. This design enables the model to outperform CDGTN and DDGCN, especially in early-stage rumor detection where causal integrity is critical. The causal attention mechanism further enhances precision on time-sensitive classes such as FR and UR, yielding state-of-the-art F1 scores on both.

(3) *Effectiveness of Feedback Refinement Mechanism.* Compared to prior models that require full recomputation upon snapshot updates (e.g., DynGCN, DA-GCN), DCAGTN employs a lightweight feedback mechanism that incrementally refines node embeddings. This not only improves computational efficiency but also stabilizes predictions over dynamic inputs, contributing to performance gains on the Weibo dataset, where rapid propagation demands real-time adaptation.

(4) *Outperforming Graph Transformer and IB-Based Methods.* While models such as PSGT and AGAD leverage graph transformers or contrastive learning to model rumor propagation, they lack explicit causal supervision or dynamic refinement. DCAGTN's integrated causal fusion and agreement-aware edge weighting allow it to surpass these methods in both precision and recall, particularly on challenging rumor types. For instance, DCAGTN improves FR-F1 on Twitter15 from 0.984 (AGAD) to 0.985 and TR-F1 from 0.947 to 0.949, validating its ability to discern fine-grained veracity features across propagation sequences.

These results collectively demonstrate the advantage of incorporating causal attention, temporal edge dynamics, and feedback-guided refinement into a unified model. DCAGTN offers a scalable, real-time solution for rumor detection, setting a new benchmark across all evaluated datasets.

To further contextualize our results, Table 4 summarizes the unique architectural features and comparative performance of DCAGTN and recent baselines, evaluated on Twitter15 and Twitter16. DCAGTN is the only model that simultaneously incorporates *causal feedback*, an explicit *alignment mechanism*, and *dual-snapshot modeling*, allowing it to capture both short-term and long-range temporal dynamics in rumor propagation.

This integrated design enables DCAGTN to achieve the highest Unverified Rumor (UR) F1-scores on both datasets, with 0.918 on Twitter15 and 0.939 on Twitter16, outperforming strong baselines such as CDGTN (0.859 / 0.931) and AGAD (0.913 / 0.915). In terms of over-

all accuracy, DCAGTN surpasses all prior methods, indicating its robustness across diverse rumor classes. These gains validate the effectiveness of our agreement-aware attention, causal fusion, and feedback refinement mechanisms in delivering real-time, interpretable rumor detection.

4.6. Ablation results

To understand the contribution of each component in DCAGTN, we performed ablation experiments by selectively disabling major modules and measuring performance degradation. Table 5 presents results across the three benchmark datasets.

- *w/o Causal Attention Module.* Removing the causal attention layer led to notable accuracy drops on all datasets, particularly 2.6% on Weibo. This confirms its critical role in preserving directional influence and capturing cause-and-effect relations within dynamic rumor spread.

- *w/o Agreement-Aware Gating.* Excluding the agreement-aware gating mechanism degraded classification performance, especially on ambiguous categories. Without this module, the model loses its ability to reweight noisy interactions using parent-child semantic similarity.

- *w/o Feedback Refinement Block.* The absence of feedback refinement reduced accuracy by up to 2.3% on Twitter16. This shows its importance in iteratively updating node relevance as context evolves during propagation.

- *w/o Alignment Mechanism.* Disabling the alignment layer caused a 3.2% drop on Weibo, reflecting its role in pruning irrelevant edges and enhancing focus on salient causal paths.

- *w/o Dual Snapshot Modeling.* Using only temporal or only causal snapshots caused consistent performance loss across datasets, confirming that both global (temporal) and local (causal) views are necessary for robust early and late-stage detection.

- *w/o GraphEncoder Blocks.* Removing both causal and non-causal graph encoders led to the largest drop, up to 11.1% on Twitter16. This validates their role in capturing structural features critical to propagation-aware classification.

Component Contributions. Table 7 quantifies the relative gain from each component. The graph encoder and causal attention provided the largest accuracy improvements, confirming their centrality to the model's success.

Table 5
Ablation study results of DCAGTN components.

Model Variant	Twitter15		Twitter16		Weibo	
	Accuracy	F1-Macro	Accuracy	F1-Macro	Accuracy	F1-Macro
w/o Causal Attention	0.894	0.865	0.878	0.846	0.887	0.873
w/o Agreement Gating	0.902	0.871	0.889	0.856	0.891	0.877
w/o Feedback Refinement	0.904	0.875	0.893	0.861	0.901	0.884
w/o Alignment Mechanism	0.889	0.859	0.874	0.848	0.905	0.881
w/o Dual Snapshots (only temporal)	0.901	0.867	0.882	0.854	0.888	0.869
w/o GraphEncoder Blocks	0.862	0.834	0.825	0.822	0.844	0.812
DCAGTN* (Full Model)	0.921	0.948	0.933	0.944	0.973	0.976

Table 6
Early-stage rumor detection performance (First six hours) across Twitter15, Twitter16, and Weibo datasets. Boldface indicates the best performance.

Model	Twitter15			Twitter16			Weibo		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
RvNN	0.718	0.724	0.721	0.734	0.739	0.737	0.902	0.907	0.905
BiGCN	0.854	0.848	0.851	0.836	0.841	0.838	0.933	0.935	0.934
D _N GCN	0.899	0.904	0.901	0.948	0.953	0.950	0.951	0.954	0.952
DA-GCN	0.934	0.931	0.933	0.935	0.937	0.936	0.936	0.934	0.935
DDGCN	0.936	0.934	0.935	0.922	0.924	0.923	0.943	0.946	0.945
CDGTN	0.891	0.888	0.890	0.917	0.919	0.918	0.967	0.969	0.968
AGAD	0.940	0.942	0.941	0.961	0.962	0.961	0.971	0.971	0.972
PSGT	0.864	0.867	0.865	0.874	0.876	0.875	0.950	0.949	0.950
FGDGNN	0.901	0.899	0.900	0.901	0.899	0.900	0.961	0.963	0.962
DCAGTN	0.943	0.945	0.944	0.965	0.967	0.957	0.979	0.973	0.975

Table 7
Accuracy gain by component removal (%).

Component Removed	T15	T16	Weibo
GraphEncoder Blocks	5.9	11.1	12.9
Causal Attention	2.7	5.5	8.6
Agreement Gating	1.9	4.4	8.2
Feedback Refinement	1.7	4.0	7.2
Alignment Mechanism	3.2	5.9	6.8
Dual Snapshots (only temp)	2.0	5.1	8.5

Table 8
Parameter sensitivity of DCAGTN on Twitter16.

Parameter	Range	Acc.	F1-M
α (alignment weight)	[0.1, 0.3, 0.5, 0.7, 1.0]	0.952–0.957	0.940–0.946
λ (temporal decay)	[0.01, 0.05, 0.1, 0.2]	0.953–0.956	0.941–0.945
N_{snap} (snapshots)	[2, 4, 6, 8]	0.951–0.958	0.940–0.947

4.7. Parameter sensitivity analysis

To evaluate the robustness of DCAGTN with respect to key hyperparameters, we conducted a parameter-sensitivity analysis over (i) the causal alignment weight α , (ii) the temporal decay factor λ , and (iii) the number of snapshots N_{snap} . Each parameter was varied while keeping all others fixed, and the average accuracy and macro F1 on the Twitter16 dataset were recorded.

Results in Table 8 show that DCAGTN remains stable across wide parameter ranges, with less than 1% variation in accuracy and macro-F1. The model performs best around $\alpha = 0.5$, $\lambda = 0.1$, and $N_{\text{snap}} = 6$, consistent with our main settings. This indicates that DCAGTN's dual-snapshot and causal-consistency mechanisms are not overly sensitive to tuning and generalize well across configurations, supporting its scalability for large and evolving social networks.

4.8. Early detection performance

We evaluated DCAGTN's capability during the early propagation stages of rumors, crucial moments for minimizing rumor spread. The model's causal-attention module, feedback mechanism, and dual-snapshot design allow it to prioritize time-sensitive interactions and adapt rapidly. To assess early detection efficacy, we measured performance within the first six hours across Twitter15, Twitter16, and Weibo datasets. Table 6 presents early-stage precision, recall, and F1-score. DCAGTN outperforms all recent baselines in early-stage rumor detection. On Twitter15, it achieves an F1-score of 0.944, surpassing AGAD (0.941), CDGTN (0.890), and FGDGNN (0.900). On Twitter16 and Weibo, it yields F1-scores of 0.957 and 0.979, respectively, reflecting strong cross-domain generalizability and early convergence. Within just six hours, DCAGTN exceeds 0.94 accuracy on Twitter15 and 0.96 on both Twitter16 and Weibo. These results confirm its robustness in time-critical rumor detection situations, where early, accurate classification is vital.

Our integration of causal attention, feedback mechanisms, and dual snapshots provides a distinct advantage in dynamically prioritizing information and capturing both immediate and longer-term propagation features. These capabilities enable DCAGTN to consistently outperform baselines in early detection situations, reinforcing its utility for real-time rumor detection systems.

4.9. Class-wise F1 breakdown

To assess DCAGTN's effectiveness in handling semantically ambiguous rumor types, we report class-wise F1-scores on Twitter15 and Twitter16, with a focus on Unverified Rumors (UR) and False Rumors (FR). Fig. 3 presents the comparative performance against representative baselines.

DCAGTN achieved substantial gains on the UR class, recording 0.918 F1 on Twitter15 and 0.939 on Twitter16, outperforming CDGTN by absolute margins of 5.9% and 0.8%, respectively. Similar improvements were observed for the FR class, where DCAGTN achieved the highest F1 across both datasets, confirming the effectiveness of agreement-aware

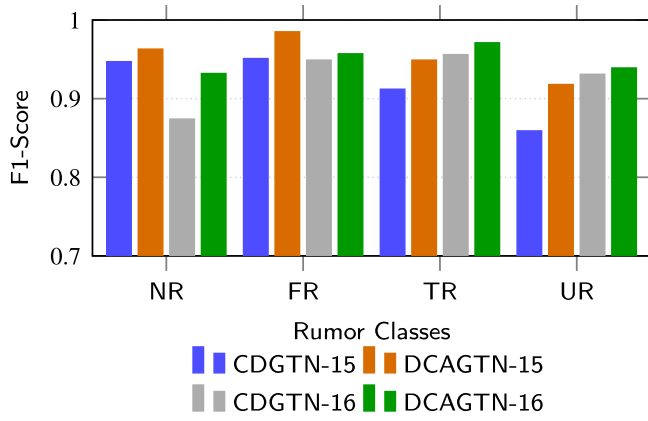


Fig. 3. Class-wise F1-score comparison on Twitter15 and Twitter16.

attention and causal refinement in accurately identifying and classifying ambiguous rumor types.

5. Conclusion

In this paper, we proposed DCAGTN, a dual-snapshot causal-attention graph Transformer designed for real-time rumor detection. By integrating causal alignment and feedback refinement into a unified attention-based graph architecture, DCAGTN captures both short-term dynamics and long-range propagation features. The dual-snapshot mechanism enables efficient updates to node representations without full graph recomputation, while causal attention ensures temporal consistency for early-stage detection. Despite its strong performance, DCAGTN currently focuses on text-based features; future work will explore integrating multimodal inputs and adaptive large language model (LLM) reasoning to enhance robustness in complex social media environments. In the future, we plan to extend DCAGTN for streaming and cross-platform events to evaluate its scalability in real-world deployments. Extensive experiments confirm the effectiveness of DCAGTN across multiple datasets, consistently outperforming strong GNN and Transformer-based baselines.

CRedit authorship contribution statement

Lazarus Kwao: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization; **Yang Yang:** Writing – review & editing, Supervision, Resources, Funding acquisition; **Wisdom Xornam Ativi:** Writing – review & editing, Writing – original draft, Methodology; **Sophyani Banaamwini Yussif:** Writing – review & editing, Writing – original draft, Validation, Methodology.

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Data availability

Public datasets (Twitter15, Twitter16, Weibo). Code and scripts available on request

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Implementation and reproducibility details

A.1. Environment and dependencies

We conducted all experiments in a reproducible Python 3.11 environment using PyTorch 2.3 and Torch Geometric 2.5. We used the transformers (v4.40) and scikit-learn (v1.5) libraries for text encoding, evaluation, and cross-validation.

A.2. Data preprocessing and snapshot construction

Raw event trees were converted into dual snapshots (sequential $S(t)$ and temporal $T(t)$) with agreement-aware and temporal-decay edge weights:

Listing 1: Snapshot construction

```
for event in dataset:
    tree = build_reply_tree(event)
    compute_agreement_scores(tree) #
    ↪ cosine([CLS]_parent, [CLS]_child)
    apply_temporal_decay(tree, lambda_) # exp(-lambda *
    ↪ Delta_t)
    snapshots = make_dual_snapshots(tree, N_snap)
    save_npz(event_id, snapshots)
```

Each saved snapshot includes node features, edge indices, edge types, normalized timestamps, and veracity labels. This process ensures consistent chronological segmentation across training and validation folds.

A.3. Dataset loading and k-fold splitting

Event-level k -fold cross-validation was used to prevent temporal leakage:

Listing 2: Event-level k-fold split

```
full_data = torch.load("samples.pt")
kf = KFold(n_splits=5, shuffle=True, random_state=42)
for train_events, val_events in kf.split(event_ids):
    train_loader = DataLoader(train_events)
    val_loader = DataLoader(val_events)
```

The collate function tokenizes each node's text using RoBERTa, re-constructs graph batches, and injects agreement-aware and temporal-decay weights for each edge.

A.4. Model forward flow

Listing 3: Forward pass (schematic)

```
def forward(X, A, R, w, prev_state):
    H_causal = GraphEncoder_c(X, A["causal"], R, w)
    H_noncausal = GraphEncoder_nc(X, A["noncausal"], R, w)
    H_fused = fuse(H_causal, H_noncausal)
    H_refined = feedback_refine(H_fused, prev_state)
    return classifier(H_refined)
```

Fusion is performed through a learnable projection combining causal and non-causal streams, followed by gated refinement that integrates the previous snapshot state.

A.5. Feedback refinement and causal alignment loss

Listing 4: Feedback refinement and causal alignment

```
# Feedback refinement
G = sigmoid(W_g @ concat(H_fused, H_prev, Delta_t) + b_g)
H_refined = G * relu(W_u @ H_fused) + (1 - G) * H_prev

# Causal alignment (supervise past -> future)
y_t = M_t * stopgrad(H_causal_prev) + (1 - M_t) *
    ↪ stopgrad(W_p @ H_fused)
L_causal = mse_loss(H_causal, y_t)
```

This design enforces chronological consistency by supervising only from past to future snapshots.

A.6. Training loop

Listing 5: Training loop

```
for epoch in range(num_epochs):
    for snapshot in snapshots:
        H_c, H_nc = model.encode(snapshot)
        y_t = build_target(H_c, H_nc, mask)
        L_causal = mse_loss(H_c, y_t)
        pred = model.classify(H_nc)
        L_total = CE_loss(pred, label) + alpha * L_causal
        L_total.backward()
        optimizer.step()
```

Training uses Adam with linear warm-up of α , gradient clipping (1.0), early stopping on validation macro-F1, and DropEdge regularization. Each fold runs for 30 epochs, with per-epoch wall-clock well under typical GPU time budgets.

Computational complexity.

$$\underbrace{2\mathcal{O}(LH(nd^2 + m'd))}_{\text{two encoders}} + \underbrace{\mathcal{O}(nd^2)}_{\text{fusion projection}} \quad (\text{A.1})$$

$$= \mathcal{O}(LH(nd^2 + m'd)),$$

where the fusion cost is lower order in practice. Over N_{snap} snapshots in a forward pass, the time is

$$\mathcal{O}(N_{\text{snap}} LH(nd^2 + m'd)). \quad (\text{A.2})$$

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